

The solutions of the system is some affine m -flat with a vector basis

$$\begin{bmatrix} \widetilde{-a_{1,r+1}} \\ \vdots \\ \widetilde{-a_{r,r+1}} \\ 1 \\ 0 \\ \vdots \end{bmatrix}, \begin{bmatrix} \widetilde{-a_{1,r+2}} \\ \vdots \\ \widetilde{-a_{r,r+2}} \\ 0 \\ 1 \\ \vdots \end{bmatrix}, \dots, \begin{bmatrix} \widetilde{-a_{1,n}} \\ \vdots \\ \widetilde{-a_{r,n}} \\ 0 \\ \vdots \\ 1 \end{bmatrix}$$

and an affine point

$$\begin{bmatrix} \tilde{b}_1 \\ \vdots \\ \tilde{b}_r \\ 0 \\ \vdots \\ 0 \end{bmatrix}.$$

In the particular case where $r = n = N$ the system has a unique solution. Start from here we consider the only this nondegenerate case. The procedure of producing the RREF of the augmented matrix $[A|\mathbf{b}]$ is written as some computer program. We refer the readers to [PTVF] for a program in C language. It is easy to show that this computation needs at least the time of order 2^N .

In this paper we try to simulate this procedure on quantum computers, and show that we could have the complexity time of order $2^{N/2}$.

The paper is organized as follows. We state our QGJE Algorithm and the Main Theorem in §2. In §3 we analyze the structure of quantum addition and explain its mechanism of work. It seems to the authors that by using the representation theory and the corresponding Fast Fourier Transformation is easy to explain the quantum adder and also quantum multiplication. In §5 we prepare the main background of quantum computing, and finally in §5 the main theorem is proved.

2 QGJE Algorithm

Step 1 Use the Grover's Search algorithm to find out the first non-zero $a_{i1} \neq 0$.

Step 2 If the search is successful, produce the first leading 1 in the first place as a_{11} , else change to the next column and repeat step 1.

Step 3 Eliminate all other entries $a_{1,1}, \dots, a_{N,1}$ in the column.

Step 4 Change N to $N - 1$, control if still $N > 0$, repeat the procedure from the step 1.

Step 5 In backward eliminate all $a_{N-1,N}, \dots, a_{1,N}$.

Step 6 Check if $N > 0$, change N to $N - 1$ and repeat the step 5.

Theorem 2.1 *In the Quantum Gauß-Jordan Elimination Algorithm one needs at least*

$$\frac{N(N-1)(2N+1)}{3} + \left\lceil \sqrt{2} \frac{(\sqrt{2})^N - 1}{\sqrt{2} - 1} \right\rceil \sim O(2^{N/2})$$

operations.

Proof of this theorem is the main goal of the paper.

3 A structural comment of Quantum Adder construction

In this section we give a brief review of quantum computers, as we need in the rest of this paper.

3.1 q -digit counting system

Let us first remind the ordinary rule of arithmetic operations. Let p be a prime and r some positive integer. Write a number in series of $q = p^r$ with coefficients in the finite field $\mathbf{F}_q$,

$$a = \sum_{i=0}^m a_i q^i, \quad b = \sum_{j=0}^n b_j q^j, \quad a_i, b_j \in \mathbf{F}_q.$$

The sum $a + b$ and the product $a.b$ are defined as usually

$$a + b = (a_0 + a_1 q + \dots) + (b_0 + b_1 q + \dots) = c_0 + c_1 q + \dots,$$

$$a.b = (a_0 + a_1 q + \dots)(b_0 + b_1 q + \dots) = d_0 + d_1 q + \dots$$

where

$$\begin{aligned}
c_0 &= a_0 + b_0(\text{mod } q), \\
c_1 &= a_1 + b_1 + [(a_0 + b_0)/q](\text{mod } q), \\
&\dots\dots\dots \\
d_0 &= a_0.b_0(\text{mod } q), \\
d_1 &= a_1.b_1 + [(a_0.b_0)/q](\text{mod } q), \\
&\dots\dots\dots
\end{aligned}$$

3.2 Qubits

A fundamental notion of quantum computers is qubit.

Definition 3.1 A *qubit* is a quantum system the Hilbert space of its quantum states is the q -dimensional complex plane $\mathbf{C}^q$, and therefore a quantum state of a qubit is a normalized complex vector $\mathbf{v} \in \mathbf{C}^q$

$$\mathbf{v} = \begin{bmatrix} v_0 \\ \vdots \\ v_{q-1} \end{bmatrix}, ||\mathbf{v}|| = |v_0|^2 + \dots + |v_{q-1}|^2 = 1.$$

Choose the standard basis of q vectors

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, |1\rangle = \begin{bmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix}, \dots, |q-1\rangle = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix}.$$

An arbitrary vector can be decomposed in this basis as a linear combination

$$\mathbf{v} = v_0|0\rangle + v_1|1\rangle + \dots + v_{q-1}|q-1\rangle.$$

Lemma 3.2 Every n -digit integer number can be written as a tensor product of n qubits

$$a = a_0 \otimes \dots \otimes a_{n-1} \in (\mathbf{C}^2)^{\otimes n}, a_i \in \mathbf{F}_q = \mathbf{Z}/q\mathbf{Z}.$$

PROOF. Every integer can be written in binary form

$$a = a_0 + a_1 2 + \dots + a_n 2^n, a_i \in \mathbf{F}_q = \{0, 1, \dots, q-1\}.$$

Therefore we have $|a_0\rangle \otimes \dots \otimes |a_n\rangle \in (\mathbf{C}^2)^n$. $\square$

Let us consider the additive groups $G = (\mathbf{F}_q)^n$. A unitary character is a continuous homomorphism $\chi : G \rightarrow \mathbf{S}^1 = \mathbf{U}(\mathbf{C})$ from the group G into the group of unitary automorphism of $\mathbf{C}$, i.e. a continuous map $\chi : G \rightarrow \mathbf{S}^1 \cong \mathbf{U}(\mathbf{C})$, such that

$$\begin{cases} \chi(x+y) &= \chi(x)\chi(y) \\ \chi(0) &= 1 \\ \|\chi(x)\| &\equiv 1. \end{cases}$$

Lemma 3.3 *Every (multiplicative) unitary character of the additive groups $G = (\mathbf{F}_q)^n$ is of the form*

$$\chi_a(x) = e^{\frac{2\pi i}{q^n} \sum_{i=1}^n a_i x_i}.$$

PROOF. The lemma is an easy exercise from the group representation theory. $\square$

It is convenient to write the number in fractional form, i.e. write

$$0.a = \frac{a_0}{q^n} + \frac{a_1}{q^{n-1}} + \dots$$

The integers modulo q^n can be written as

$$\bar{k} \mapsto e^{2\pi i 0.k}, 0.k = k/q^n,$$

for $k = 0, 1, \dots, q^n - 1$. Let us consider the Haar measure $\mu(k) = \frac{k}{q^n}$. Then we have the natural *Fast Fourier Transform (FFT)*.

Definition 3.4 The transformation

$$FFT : |y\rangle \mapsto \frac{1}{2^n} \sum_{k=0}^{q^n-1} e^{2\pi i y_k \frac{k}{q^n}} |k\rangle$$

is called the *Fast Fourier Transform*

3.3 Quantum addition

By using FFT we have a 1-1 correspondence

$$\begin{aligned} a &\xrightarrow{FFT} e^{2\pi i 0.a_n a_{n-1} \dots a_0} \\ b &\xrightarrow{FFT} e^{2\pi i 0.b_n b_{n-1} \dots b_0} \end{aligned}$$

Definition 3.5 The sum of two numbers a and b is defined as follows.

$$\begin{aligned} a &\xrightarrow{FFT} e^{2\pi i 0.a_n \dots a_0} \xrightarrow{\text{add } 0.0\dots 0b_0} e^{2\pi i 0.a_n a_{n-1} \dots a_1 (a_0 + b_0)} \xrightarrow{\text{add } 0.0\dots 0b_1} \dots \\ &\dots \xrightarrow{\text{add } 0.b_n} e^{2\pi i 0.(a_n + b_n) \dots (a_0 + b_0)} \xrightarrow{FFT \text{ inverse}} a + b \end{aligned}$$

Remark 3.6 The reduction (memorize $[(a_i + b_i)/q]$ to $a_{i+1} + b_{i+1}$ is the phase transition with the matrix

$$\begin{bmatrix} 1 & 0 \\ 0 & e^{2\pi i \frac{[(a_i + b_i)/q]}{q^{i-1}}} \end{bmatrix}$$

3.4 Quantum gates

Let us recall some fundamental gates in quantum computing.

3.4.1 Hadamard gate

Definition 3.7 The 1-qubit Hadamard gate $-\boxed{H}-$ is defined by the matrix

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}.$$

The Hadamard gates acts on qubits as follows

$$|x\rangle \xrightarrow{-\boxed{H}-} (-1)^x |x\rangle + |1 - x\rangle.$$

3.4.2 Phase gate

Definition 3.8 The phase 2-qubit gate $-\boxed{\Phi}-$ is defined by the matrix

$$\Phi = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 0 \\ 0 & e^{i\Phi} \end{bmatrix}.$$

it produces an action

$$|x\rangle \xrightarrow{-\boxed{\Phi}-} e^{i\Phi} |x\rangle.$$

3.4.3 CNOT(XOR) gate

Definition 3.9 The 2-qubit C-NOT (XOR) gate is defined by the matrix

$$C = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}.$$

Action:

$$\begin{cases} |x\rangle & \rightarrow |x\rangle \\ |y\rangle & \rightarrow |x \oplus y\rangle, \end{cases}$$

where $x \oplus y := x + y \bmod(2)$.

3.4.4 Unitary gates

Definition 3.10 The unitary gate $\boxed{U}$ is defined by a unitary matrix U ,

$$U = \begin{bmatrix} 1 & 0 \\ 0 & U \end{bmatrix}.$$

3.4.5 Problem of quantum computing

One of the major result concerning this problem is the following classification result.

Theorem 3.11 (Universal gates) One-qubit gates, CNOT gate, phase gate and one of unitary gate generate all the other gates.

4 Quantum Grover's Search Algorithm

4.0.6 Black-box (Query) problem

The problem is to compute some Boolean function of type

$$f : \{0, 1\}^n \rightarrow \{0, 1\}.$$

4.1 Deutsch's original problem

The problem is to check whether the above defined Boolean function is balance: the number of values 0's equals to the numbers of values 1's, or constant.

4.2 Search problem

Given a Boolean function $f : \{0, 1\}^n \rightarrow \{0, 1\}$, defined by $f_k(x) = \delta_{xk}$. Find k .

4.3 Deutsch's scheme $\boxed{\text{H}} - \boxed{\text{f}} - \boxed{\text{H}} -$

The Deutsch's scheme produces the effects:

$$|x\rangle(|0\rangle - |1\rangle) \xrightarrow{\boxed{\text{H}} - \boxed{\text{f}} - \boxed{\text{H}} -} \underbrace{(-1)^{f(x)}|x\rangle}_{\text{measurement}} (|0\rangle - |1\rangle).$$

In this scheme, the second qubit is of no interest, while the first qubit has state

$$(-1)^{f(0)}|0\rangle + (-1)^{f(1)}|1\rangle = \begin{cases} \pm(|0\rangle + |1\rangle), & \text{if } f = \text{const} \\ \pm(|0\rangle - |1\rangle), & \text{if } f = \text{balanced} \end{cases}$$

4.4 Quantum Grover's Search Algorithm

Scheme: $\boxed{f_k} - \boxed{\text{H}} - \boxed{f_0} - \boxed{\text{H}} -$.

This algorithm has special effect: Choose the state

$$|S\rangle = \frac{1}{2^{N/2}} \sum_{i=1}^{N-1} |i\rangle,$$

and produce the Grover's iterate

$$|\psi\rangle(|0\rangle - |1\rangle) \xrightarrow{\boxed{f_k} - \boxed{\text{H}} - \boxed{f_0} - \boxed{\text{H}} -} |\text{measurement}\rangle(|0\rangle - |1\rangle).$$

This iterate does nothing to any basis element $|i\rangle$ except for $|k\rangle$ is changing into $-|k\rangle$, i.e. reflection in the hyperplane perpendicular to the plane

$$H|0\rangle = |S\rangle = \frac{1}{2^{N/2}} \sum_{i=1}^{N-1} |i\rangle,$$

i.e. rotation in the plane generate by $|k\rangle$ and $|S\rangle$ an the Grover's iterate is a rotation of twice the angle from $|k^\perp\rangle$ to $|S^\perp\rangle$.

5 Proof of the Main Theorem

STEP 1. Let us consider the first column $\begin{bmatrix} a_{11} \\ \vdots \\ a_{N1} \end{bmatrix}$. Each a_{i1} is either zero or

non-zero. We do find the first non-zero value. In classical computation we need to check all possible values of column entries and we need 2^N operations. In quantum Grover's Search Algorithm we need to proceed $2^{N/2}$ operations. More precisely, choose a state

$$|S\rangle = 2^{-N/2} \sum_{i=0}^{N-1} |i\rangle$$

and produce the Grover's iterate

$$|\Psi\rangle(|0\rangle - |1\rangle) \xrightarrow{[f_k] \text{---} [H] \text{---} [f_0] \text{---} [H]} |\text{measurement}\rangle(|0\rangle - |1\rangle).$$

This iteration does nothing to any basis element except for $|k\rangle$ is changing into $-|k\rangle$, i.e. reflection in the hyperplane perpendicular to plane

$$H|0\rangle = 2^{-N/2} \sum_{i=0}^{N-1} |i\rangle = |S\rangle,$$

i.e. a rotation in the plane generated by $|k\rangle$ and $|S\rangle$, and the desired state $|k\rangle$ exactly is Φ , the iterate should be repeated m times with

$$(2m + 1)\Phi \approx \frac{\pi}{2} \text{ or } m \approx \frac{\pi}{4\Phi} - \frac{1}{2}.$$

Since

$$\frac{1}{2^{N/2}} = \sin \Phi \approx \Phi, \text{ we have } m \approx \frac{\Phi}{4} 2^{N/2}.$$

This estimate is optimal in the sense that any another quantum algorithm for searching an unstructured database must take time at least of order $O(2^{N/2})$.

Suppose we have found a non-zero element and denote it (the first nonzero element) by a_{i1}

STEP 2. Produce the first modified row

$$\mathbf{r}_1 = [1, \frac{a_{i2}}{a_{i1}}, \dots, \frac{a_{iN}}{a_{i1}}].$$

For this we need at least $N - 1$ multiplications with $\frac{1}{a_{i1}}$. Register the resulted row as the first row.

STEP 3.

- Multiply the new first row with $-a_{j1}$. For this we need $N - 1$ operations of multiplication and
- Add the result to the j^{th} row. For this we need another $N - 1$ operations.

Totally in this step we need $2(N - 1)^2$ arithmetic operations.

STEP 4. Now change N to $N - 1$ (need 1 subtraction) and control for the value $N > 0$. We need one CNOT gate.

Loop back to the first step produce summation from 1 to N .

STEPS 5 AND 6. Eliminating backward the numbers $a_{N-1,N}, \dots, a_{1,N}$, we need $N - 1$ additions and one CNOT gate to control and change N into $N - 1$.

ALL STEPS. Totally in all steps we need at least

$$\sum n = 1^N (2^{n/2} + 2(n - 1)^2 + 2(n - 1) + 1 + 1)$$

operations.

REDUCTION. The following lemma is trivial

Lemma 5.1 (Combinatorial Lemma)

$$\sum_{n=1}^N n^2 = \frac{N(N + 1)(2N + 1)}{6}.$$

FINISH THE PROOF OF THE MAIN THEOREM Denote the integral part of a real number a by $[a]$. Finally we have

$$\begin{aligned} \sum_{n=1}^N (2^{n/2} + 2(n - 1)^2 + 2(n - 1) + 1 + 1) &\approx \\ 2 \sum_{n=1}^N n + 2 \sum_{n=1}^N (n - 1)^2 + \left[\sum_{n=1}^N \sqrt{2}^n \right] &= \\ 2 \frac{N(N + 1)}{2} + 2 \frac{(N - 1)N(2N - 1)}{6} + \left[\sqrt{2} \frac{\sqrt{2}^N - 1}{\sqrt{2} - 1} \right] &= \\ \frac{(N - 1)N(2N + 1)}{3} + \left[\sqrt{2} \frac{\sqrt{2}^N - 1}{\sqrt{2} - 1} \right] &\approx O(2^{N/2}). \end{aligned}$$

Corollary 5.2 *For $N \gg 1$, the number of operations needed to produce the RREF of a square matrix using QGJE Algorithm is of order $O(2^{N/2})$.*

In the next paper [DG] we apply this computation to simulating of accounting principles on quantum computers.

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